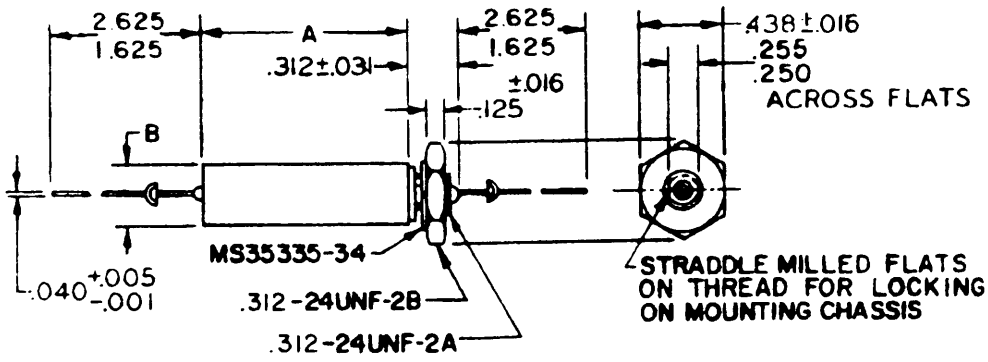
MIL-C-11693/7D
8 June 1984
SUPERSEDING
MIL-C-0011693/7C(USAF)
20 September 1974
MIL-C-11693/7A
8 October 1971

MILITARY SPECIFICATION SHEET

CAPACITORS, ~~FEED THROUGH, RADIO-INTERFERENCE REDUCTION,~~ DC (HERMETICALLY SEALED IN METAL CASES) ESTABLISHED AND NON-ESTABLISHED RELIABILITY STYLES CZ23, CZ24, CZR23 AND CZR24

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-11693.



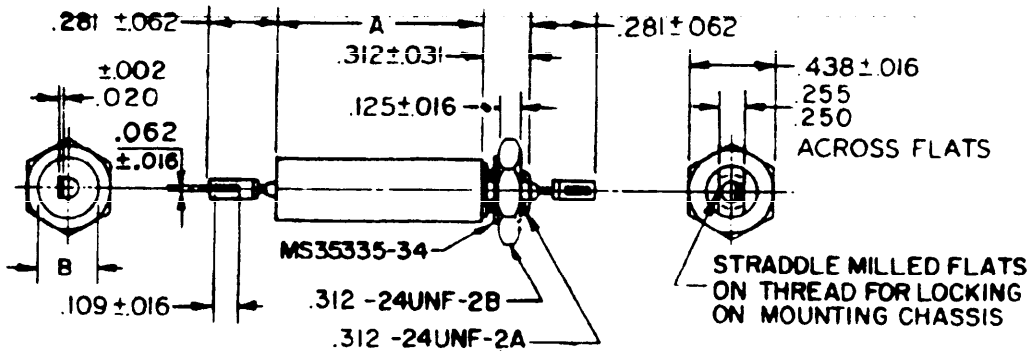
STYLES CZ23 AND CZR23

FIGURE 1. Styles CZ23, CZ24, CZR23 and CZR24 capacitors.

Ⓓ denotes changes

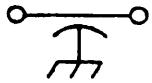
FSC 5910

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STYLES CZ24 AND CZR24

CIRCUIT DIAGRAM



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.001	0.03	.040	1.02	.281	7.14	1.00	25.4
.002	0.05	.05	1.3	.312	7.92	1.12	28.5
.005	0.13	.062	1.57	.400	10.16	1.38	35.1
.015	0.38	.109	2.77	.438	11.13	1.625	41.28
.016	0.41	.125	3.18	.562	14.27	1.88	47.8
.020	0.51	.234	5.94	.670	17.02	2.12	53.9
.031	0.79	.250	6.35	.75	19.1	2.38	60.5
		.255	6.48	.88	22.4	2.625	66.68

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.031 (0.79 mm).
4. Threaded neck to mounting shoulder shall be perpendicular with ±3°.

FIGURE 1. Styles CZ23, CZ24, CZR23, and CZR24 capacitors - Continued.

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TABLE I. Type designation, dimensions, and electrical characteristics.

Type 1/ designation	Rated voltage	Nominal capacitance	Rated current	Characteristics	Dimensions A ± .05	Dimensions B ± .015
	Volts, dc	µF	Amperes			
CZ-2-B-B473-	100	0.047	10	K, E	.75	.400
CZ-2-B-B104-	100	0.1	10	K, E	.88	.400
CZ-2-B-B224-	100	0.22	10	K, E	1.38	.400
CZ-2-B-B474-	100	0.47	10	K, E	1.38	.562
CZ-2-B-B105-	100	1.0	10	K, E	1.88	.670
CZ-2-B-B205-	100	2.0	10	K, E	2.12	1.00
CZ-2-B-C473-	200	0.047	10	K, E	.88	.400
CZ-2-B-C104-	200	0.1	10	K, E	1.12	.400
CZ-2-B-C224-	200	0.22	10	K, E	1.12	.562
CZ-2-B-C474-	200	0.47	10	K, E	1.88	.562
CZ-2-B-C105-	200	1.0	10	K, E	2.12	.750
CZ-2-B-E104-	400	0.1	10	K, E	1.12	.562
CZ-2-B-E224-	400	0.22	10	K, E	1.88	.562
CZ-2-B-E474-	400	0.47	10	K, E	2.12	.750
Ⓓ CZ-2-B-F472-	600	0.0047	10	K, E	.75	.400
CZ-2-B-F103-	600	0.01	10	K, E	.75	.400
CZ-2-B-F473-	600	0.047	10	K, E	1.38	.400
CZ-2-B-F104-	600	0.1	10	K, E	1.38	.562
CZ-2-B-F224-	600	0.22	10	K, E	1.88	.670
CZ-2-B-F474-	600	0.47	10	K, E	2.38	.750

Ⓓ 1/ Complete type designation will include the following:
1st dash - Symbol "R" (for sytles CZR23 and CZR24) or dash will be deleted (for styles CZ23 and CZ24).
2nd dash - Applicable style number.
3rd dash - Applicable characteristic (K or E).
4th dash - Applicable failure rate level symbol (CZR23 and CZR24 only) or dash will be deleted (for styles CZ23 and CZ24).

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Dielectric: Paper, or paper-plastic.

Characteristic: See table I.

Temperature range: Characteristic E -55°C to +85°C.
Characteristic K -55°C to +125°C.

Rated voltage: See table I.

Rated current: See table I.

Capacitance: See table I.

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Insertion loss (abbreviated): MIL-STD-220.

Measurements shall be made at the following frequencies for the capacitance values specified.

Nominal capacitance (µF)	Frequency (MHz)
0.0047	400 and 1,000
0.01	400 and 1,000
0.047	400 and 1,000
0.1	400 and 200
0.22	400 and 100
0.47	400 and 80
1.0	400 and 50
2.0	400 and 50

DC resistance: The dc resistance shall not exceed 0.01 ohm.

Vibration, high frequency: Method 204 of MIL-STD-202 condition A (10g).

Thermal shock: Method 107 of MIL-STD-202.

IR at 25°C - Not less than 30 percent of the value specified in the initial requirement.

Capacitance - Change not more than ±5 percent from the initial measurement.

Moisture resistance: Method 106, of MIL-STD-202.

IR at 25°C - Not less than 30 percent of the initial requirement.

Capacitance - Change not more than ±5 percent from the initial measurement.

Life (accelerated): Method 108, of MIL-STD-202.

IR at 25°C - Not less than 30 percent of the value specified in the initial requirement.

Capacitance - Change not more than ±5 percent from the initial measured value.

Custodians
Army - ER
Navy - EC
Air Force - 11

Review activities:
Air Force - 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Army - ER

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